Distro Layers

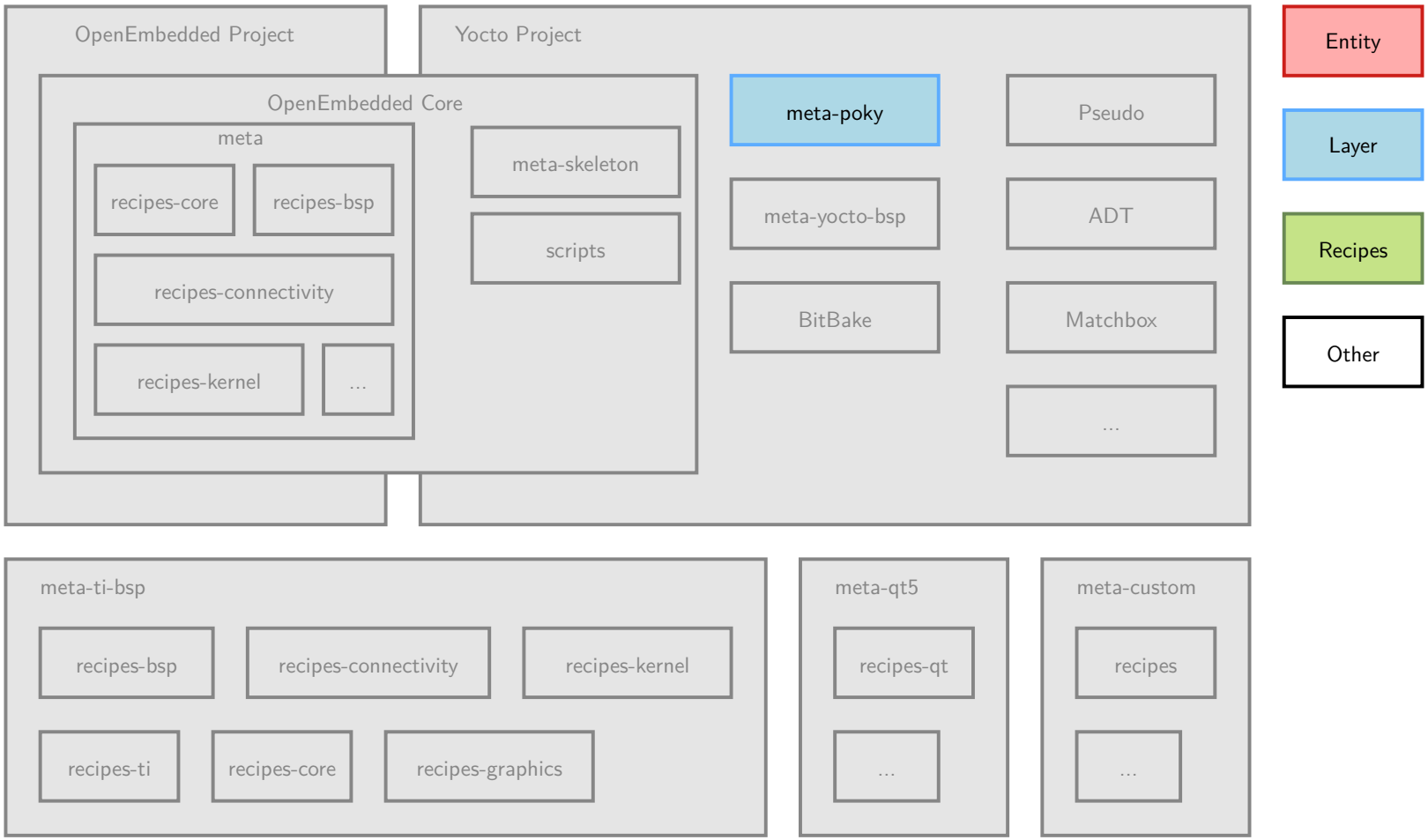


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Distro Layers

Distro layers





Distro layers

- You can create a new distribution by using a Distro layer.
- This corresponds to the settings that have an impact on your packages. You can also decide to use Musl or Glibc, Wayland or X11, systemd or sysvinit...
- A distribution layer allows to change the defaults that are provided by `openembedded-core` or `poky`.
- It is useful to distribute changes that have been made in `local.conf`
- Note: Poky is a rather bloated distribution, mainly meant to be used for testing. It's not necessarily a good starting point to optimize the root filesystem for your own platform.



Best practice

- A distro layer is used to provide policy configurations for a custom distribution.
- It is a best practice to separate the distro layer from the custom layers you may create and use.
- It often contains:
  - Distro configuration files.
  - Specific classes (for example to sign images)
  - Distribution specific recipes: initialization scripts, splash screen...



Creating a Distro configuration file

- The configuration file for the distro layer is `conf/distro/<distro>.conf`
- This file must define the DISTRO_NAME variable.
- You can also use all the `DISTRO_*` variables.
- Use `DISTRO = "<distro>"` in `local.conf` to use your distro configuration.

```
DISTRO_NAME = "My Custom Distro"
DISTRO_VERSION = "1.0"
MAINTAINER = "..."

DISTRO_FEATURES = "sysvinit ipv4 ipv6 wifi zeroconf usbgadget usbhost pni-names"
```



DISTRO_FEATURES

- Lists the features the distribution will enable (SSL, WiFi, Bluetooth...).
- As for MACHINE_FEATURES, this is used by package recipes to enable or disable functionalities.
- For example, the `bluetooth` feature:
  - Asks the `bluez` daemon to be built and added to the image.
  - Enables bluetooth support in `ConnMan`.
- COMBINED_FEATURES provides the list of features that are enabled in both MACHINE_FEATURES and DISTRO_FEATURES.



Toolchain selection

- The toolchain selection is controlled by the TCMODE variable.
- It defaults to `"default"`.
- The `conf/distro/include/tcmode-${TCMODE}.inc` file is included.
  - This configures the toolchain to use by defining preferred providers and versions for recipes such as `gcc`, `binutils`, `*libc`...
- The providers' recipes define how to compile or/and install the toolchain.
- Toolchains can be built by the build system or external (rarely used because toolchains are fast to rebuild thanks to the shared state cache).



Sample files

- A distro layer often contains `sample files`, used as templates to build key configurations files.
- Example of `sample files`:
  - `bblayers.conf.sample`
  - `local.conf.sample`
- In `Poky`, they are in `meta-poky/conf/templates/default/`.
- The TEMPLATECONF environment variable controls where to find the samples and can be exported before sourcing `oe-init-build-env`.
- `##OEROOT##` can be used in `bblayers.conf.sample` and will be replaced by the path to the directory containing the `oe-init-build-env` script, after `oe-init-build-env` is sourced.
- `bitbake-layers save-build-conf` can be used to save the current configuration.